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TaoTerm: Composable Objective Function Terms

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Current state of TAO

- Currently, objectives to **TAO** optimizer can be set via callback functions

TAO with callbacks example

```
1
2  static int MyObjAndGrad(Tao tao, Vec X, PetscReal *f, Vec G, void *ptr) {
3    //... user obj and gradient evaluation
4
5    //If you want to add regularzation to your problem,
6    //you need to edit this code manually
7    return 0;
8  }
9
10 //Simplified code for illustration purpose
11 int main(int argc, char **argv) {
12   Vec X;
13   Tao tao;
14
15   TaoCreate(MPI_COMM_WORLD, &tao);
16   TaoSetType(tao, TAOCG);
17   TaoSetObjectiveAndGradient(tao, NULL, MyObjAndGradient, &context);
18   TaoSetFromOptions(tao);
19   TaoSolve(tao);
20
21   return 0;
22 }
```

Current state of TAO

- Currently, objectives to **TAO** optimizer can be set via callback functions
- If you want to add regularization to your problem, you need to **manually edit** the callback function code
- Not composable, no clear separation between problem, instance data, and solution algorithm

What is TaoTerm?

- **TaoTerm** is a new PETSc object that enables **composable** objective functions for TAO optimization solvers
- TaoTerm is **real-valued**, **parametric function** : $f(x; p)$, where x is solution vector, and p is parameter vector
- Each TaoTerm is equipped with **objective, gradient, and/or Hessian** evaluation routines
- TaoTerm objects can be easily added to main Tao object
- (Note: Not on main branch yet. Hopefully soon)

Currently Supported TaoTerm Types

- `TAOTERMHALFL2SQUARED`: $\frac{1}{2}\|x-p\|_2^2$
- `TAOTERML1`: $\|x-p\|_1$
- `TAOTERMQUADRATIC`: $\frac{1}{2}(x-p)^T W(x-p)$
- `TAOTERMSHELL`: shell
- `TAOTERMSUM`: internal object to sum all the terms together

How to use TaoTerm - Codes

```
1  //signature
2  PetscErrorCode TaoAddTerm(Tao , const char *, PetscReal, TaoTerm, Vec, Mat);
3  ...
4
5  // tao = f(x) = user-defined
6  TaoSetObjectiveAndGradient(tao, ...);
7
8  // reg_term = g(x) = |x|_1
9  TaoTermCreate(comm, &reg_term);
10 TaoTermSetType(reg_term, TAOTERML1);
11
12 // Just simply add term
13 // tao = f(x) + 0.1 * |x-p|_1
14 //     = f(x) + 0.1 * g(x;p)
15 TaoAddTerm(tao, "reg_", 0.1., reg_term, param, NULL);
16
17 // third_term = h(x) = 0.5*|x|_2^2
18 TaoTermCreate(comm, &third_term);
19 TaoTermSetType(third_term, TAOTERMHALFL2SQUARED);
20
21 // You can also map your terms with matrix
22 // tao = f(x) + 0.1*g(x;p) + 0.5 * |Ax|_2^2
23 //     = f(x) + 0.1*g(x;p) + 0.5 * h(Ax)
24 TaoAddterm(tao, "third_", 1., third_term, NULL, Amat);
```

How to use TaoTerm - command lines

```
//you can also use command options to add terms on fly
-tao_add_terms lasso_ -lasso_tao_term_type l1 -tao_term_sum_lasso_scale 0.1

// you can turn off certain evaluations of each terms
// L1 term is now only used for objective evaluation
// note that SUM is automatically created
-tao_term_sum_lasso_mask gradient,hessian

//can also control memory type, fine hessian controls, etc
-tao_add_terms ridge_ -ridge_tao_term_type halfl2squared
-tao_term_ridge_hessian_mat_type {aij,dense,....}
-tao_term_ridge_hessian_pre_is_hessian {0,1}
-tao_term_ridge_gradient_used_fd {0,1}
-tao_term_ridge_hessian_used_fd {0,1}
```